# Delayed Bivariate Martingale DKW-Type Bounds for Online Pairwise Event Probabilities

Working technical note

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**Abstract**

We study online estimation of a bivariate lower-orthant probability when each observation is revealed only after a fixed delay and the observations may be adaptively dependent. The target is the running average of origin-conditional bivariate CDFs,

$$p_k(a,b) = \mathbb{P}(B_{k,1} \le a,\, B_{k,2} \le b \mid \mathcal{F}_k),$$

rather than an i.i.d. population CDF. A delay-$d$ staggering construction turns the delayed observations into $d$ martingale-difference streams. On a finite bivariate query grid, conditional Hoeffding concentration then yields a single high-probability event that is simultaneous over grid points, residue classes, edges, and online sample sizes. Aggregating the staggered streams gives the explicit delay-aware DKW-type width

$$\sqrt{\frac{d}{2N_t}\log\!\left(\frac{\pi^2 dQN_t^2}{3\delta}\right)},$$

where $N_t$ is the number of fully resolved origins by physical time $t$. When $N_t \asymp t$, this is

$$O\left(\sqrt{\frac{d}{t}\log\!\left(\frac{dQt^2}{\delta}\right)}\right).$$

If the conditional bivariate CDF is stable, this is a confidence band for the common CDF. If the CDF drifts, the same statistical term combines with an explicit tracking-bias term; a finite-horizon moving-window version is also derived. We additionally study adaptive convex aggregation of several memory scales. A simultaneous confidence event remains valid after data-dependent aggregation, while a residue-wise fixed-share update yields interval-wise adaptive regret for Brier loss and therefore an oracle inequality for average squared CDF estimation error. A cumulative lower-orthant variation budget then turns this oracle inequality into an explicit nonstationary estimation bound, with the meta-learner adapting to the unknown variation scale through its choice of memory. The note is deliberately finite-grid: under arbitrary adapted data, a continuum-uniform martingale DKW extension is not automatic, and sequential rather than classical empirical-process complexity is the relevant object.

## 1 Motivation and scope

Many online problems require estimating a probability of simultaneous events. A useful representation is

$$p_k(a,b) := \mathbb{P}(B_{k,1} \le a,\ B_{k,2} \le b \mid \mathcal{F}_k)\,, \tag{1}$$

where $B_k = (B_{k,1}, B_{k,2})$ is generated at origin $k$ but is only fully observed after a delay. In the trajectory-risk application motivating this note, $B_k$ is a pair of critical conformal levels and $p_k(a, b)$ is the joint failure probability of two forecast horizons at marginal risk levels $(a, b)$.

The classical Dvoretzky–Kiefer–Wolfowitz (DKW) inequality concerns an i.i.d. empirical CDF. Here we instead want three features simultaneously:

1. *adaptive dependence*: no independence across origins is assumed;

2. *bivariate queries*: the object of interest is a lower-orthant CDF;

3. *delayed observation*: $B_k$ is unavailable at the time its conditional law in (1) is defined.

The main result below is a finite-grid martingale analogue of a bivariate DKW bound. It does not require i.i.d. data, stationarity, mixing, or independence between edges. The price is that the theorem controls a finite collection of lower-orthant queries rather than the entire continuum.

## 2 Relation to martingale empirical-process results

Rakhlin, Sridharan, and Tewari study the dependent empirical process

$$\frac{1}{n}\sum_{i=1}^{n}\left(f(Z_i) - \mathbb{E}[f(Z_i) \mid \mathcal{A}_{i-1}]\right) \tag{2}$$

uniformly over a function class $f \in \mathcal{H}$, using sequential symmetrization, sequential covering numbers, and sequential fat-shattering complexity [1]. Their Lemmas 10–11 give high-probability bounds of precisely this martingale-empirical-process form.

Yin, Agrawal, and Zeevi use these results to state a martingale version of the univariate DKW inequality for adaptive reported values [2]. Their Lemma 1 compares the empirical CDF to the running average conditional CDF, which is the same type of target used in this note.

There is, however, an important verification point. Rakhlin et al. emphasize that the relevant complexity under arbitrary adapted dependence is *sequential* complexity, and they explicitly note that classical threshold classes need not satisfy unrestricted sequential uniform convergence. Consequently, extending a continuum-uniform univariate martingale DKW statement to bivariate lower orthants cannot be justified merely from ordinary VC dimension or ordinary fat-shattering dimension.

For this reason, the main theorem here avoids a continuum-complexity claim. We work on a finite query grid and prove the concentration result directly from conditional Hoeffding inequalities. The result is therefore self-contained once the delayed martingale filtration is established.

Mineiro and Howard provide another complementary viewpoint: time-uniform and value-uniform confidence bands for the *running averaged conditional CDF* under nonstationarity [3]. This reinforces an important interpretation used below: absent additional temporal regularity, the natural martingale target is an average of conditional laws, not an arbitrary current conditional law.

## 3 Delayed bivariate observations

Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration representing information available at physical time $t$. Fix one pairwise process with deterministic delay $d \in \mathbb{N}$. For every origin $k \geq 1$, let

$$B_k = (B_{k,1}, B_{k,2}) \in \mathbb{R}^2$$

be measurable with respect to $\mathcal{F}_{k+d}$. Thus the pair generated at origin $k$ is fully observed by physical time $k+d$.

Let

$$\mathcal{Q} = \mathcal{G}_1 \times \mathcal{G}_2$$

be a finite set of lower-orthant queries, with $Q := |\mathcal{Q}|$. In the common square-grid case, $\mathcal{G}_1 = \mathcal{G}_2 = \mathcal{G}$ and $Q = M^2$ where $M = |\mathcal{G}|$.

For $g = (a, b) \in \mathcal{Q}$, define

$$X_k(g) := \mathbf{1}\{B_{k,1} \le a,\ B_{k,2} \le b\}, \tag{3}$$

$$p_k(g) := \mathbb{E}[X_k(g) \mid \mathcal{F}_k]. \tag{4}$$

The function $g \mapsto p_k(g)$ is the origin-$k$ conditional bivariate CDF restricted to the query set $\mathcal{Q}$.

### 3.1 Why delay breaks the naive martingale indexing

The centered variable

$$X_k(g) - p_k(g)$$

has mean zero conditional on $\mathcal{F}_k$, but it is not observed until time $k+d$. If all origins are indexed consecutively, then the filtration immediately preceding the reveal of $X_k(g)$ contains information strictly beyond $\mathcal{F}_k$. Thus the sequence is not automatically a martingale-difference sequence with respect to the reveal-time filtration while retaining $p_k(g)$ as the target.

The following staggering construction fixes this exactly.

## 4 Staggered delayed martingale streams

For residue $c \in \{1, \dots, d\}$, define the origins

$$k_{c,i} := c + (i-1)d, \qquad i = 1, 2, \dots. \tag{5}$$

Define the staggered filtration

$$\mathcal{H}_{c,i} := \mathcal{F}_{c+id}. \tag{6}$$

Then

$$\mathcal{H}_{c,i-1} = \mathcal{F}_{k_{c,i}},$$

and $B_{k_{c,i}}$ is measurable with respect to $\mathcal{H}_{c,i}$ because $k_{c,i} + d = c + id$.

For $g \in \mathcal{Q}$, write

$$X_{c,i}(g) := X_{k_{c,i}}(g), \tag{7}$$

$$p_{c,i}(g) := p_{k_{c,i}}(g), \tag{8}$$

$$D_{c,i}(g) := X_{c,i}(g) - p_{c,i}(g). \tag{9}$$

**Lemma 1** (Delayed martingale difference)**.** *For every residue $c$ and query $g \in \mathcal{Q}$, $(D_{c,i}(g))_{i \ge 1}$ is a martingale-difference sequence with respect to $(\mathcal{H}_{c,i})_{i \ge 0}$, and $D_{c,i}(g) \in [-1, 1]$.*

*Proof.* The variable $X_{c,i}(g)$ is $\mathcal{H}_{c,i}$-measurable. Moreover,

$$\mathbb{E}[X_{c,i}(g) \mid \mathcal{H}_{c,i-1}] = \mathbb{E}[X_{k_{c,i}}(g) \mid \mathcal{F}_{k_{c,i}}] = p_{c,i}(g).$$

Hence $\mathbb{E}[D_{c,i}(g) \mid \mathcal{H}_{c,i-1}] = 0$. Since $X_{c,i}(g) \in \{0, 1\}$ and $p_{c,i}(g) \in [0, 1]$, the increment lies in $[-1, 1]$. □

The staggering is therefore a filtration device. It is not an independence assumption and it does not assert that different residue classes are independent.

## 5 A time-uniform finite-grid martingale DKW-type bound

For one residue class, define the empirical and average conditional CDFs

$$\widehat{p}_{c,n}(g) := \frac{1}{n}\sum_{i=1}^{n} X_{c,i}(g), \tag{10}$$

$$\bar{p}_{c,n}(g) := \frac{1}{n}\sum_{i=1}^{n} p_{c,i}(g). \tag{11}$$

**Theorem 2** (Anytime delayed bivariate grid bound)**.** *Fix $\delta \in (0,1)$. For $n \geq 1$, define*

$$r_n(\delta) := \sqrt{\frac{1}{2n}\log\left(\frac{\pi^2 dQn^2}{3\delta}\right)}. \tag{12}$$

*Then, with probability at least $1-\delta$,*

$$|\widehat{p}_{c,n}(g) - \bar{p}_{c,n}(g)| \leq r_n(\delta) \tag{13}$$

*simultaneously for every residue $c \in \{1,\dots,d\}$, every query $g \in \mathcal{Q}$, and every sample size $n \geq 1$.*

*Proof.* Fix $c$, $g$, and $n$. Conditional Hoeffding concentration for the bounded martingale-difference sequence in Lemma 1 gives

$$\mathbb{P}(|\widehat{p}_{c,n}(g) - \bar{p}_{c,n}(g)| > x) \leq 2e^{-2nx^2}.$$

With $x = r_n(\delta)$, the right-hand side equals

$$\frac{6\delta}{\pi^2 dQn^2}.$$

Union bounding over $d$ residues and $Q$ grid queries, and then over all $n \geq 1$, gives total failure probability at most

$$dQ\sum_{n=1}^{\infty}\frac{6\delta}{\pi^2 dQn^2} = \delta.$$

□

**Remark 3** (One confidence parameter)**.** *The proof uses multiplicity mathematically, but the algorithm requires only one user-specified level $\delta$. The factors $d$ and $Q$ appear logarithmically inside a single global radius. There is no need to maintain separate confidence levels $\delta_{c,g}$.*

## 6 Aggregating residues into one bivariate CDF estimate

At physical time $t$, let $n_{t,c}$ be the number of fully resolved origins in residue $c$, and define

$$N_t := \sum_{c=1}^{d} n_{t,c}.$$

With one origin issued at every physical time and no missing observations, $N_t = (t-d)_+$ up to the initial indexing convention.

Define the aggregate empirical CDF and its aggregate conditional target by

$$\widehat{p}_t(g) := \frac{1}{N_t} \sum_{c=1}^{d} \sum_{i=1}^{n_{t,c}} X_{c,i}(g), \tag{14}$$

$$\bar{p}_t(g) := \frac{1}{N_t} \sum_{c=1}^{d} \sum_{i=1}^{n_{t,c}} p_{c,i}(g). \tag{15}$$

When $N_t = 0$, the estimate is left undefined.

**Corollary 4** (Edge-level delayed CDF band)**.** *On the event of Theorem 2, for every $t$ with $N_t > 0$ and every $g \in \mathcal{Q}$,*

$$|\widehat{p}_t(g) - \bar{p}_t(g)| \le E_t, \qquad E_t := \sum_{c:n_{t,c}>0} \frac{n_{t,c}}{N_t} r_{n_{t,c}}(\delta). \tag{16}$$

*Moreover,*

$$\boxed{E_t \le \sqrt{\frac{d}{2N_t} \log\left(\frac{\pi^2 d Q N_t^2}{3\delta}\right)}}. \tag{17}$$

*When one origin is issued at every physical time and no observations are missing, $N_t = (t-d)_+$ up to the indexing convention. Therefore the preceding bound can be written explicitly as*

$$E_t \le \sqrt{\frac{d}{2(t-d)} \left[2\log(t-d) + \log\left(\frac{\pi^2 d Q}{3\delta}\right)\right]}, \qquad t > d,$$

*and hence, when $t \gg d$,*

$$E_t = O\left(\sqrt{\frac{d}{t} \log\left(\frac{dQt^2}{\delta}\right)}\right).$$

*Proof.* The first inequality follows by writing the aggregate difference as the weighted sum of the residue-level differences. For the second, set

$$L_t := \log\left(\frac{\pi^2 d Q N_t^2}{3\delta}\right).$$

Since $n_{t,c} \le N_t$, $r_{n_{t,c}}(\delta) \le \sqrt{L_t/(2n_{t,c})}$. Therefore

$$\begin{aligned} E_t &\le \frac{\sqrt{L_t/2}}{N_t} \sum_{c:n_{t,c}>0} \sqrt{n_{t,c}} \\ &\le \frac{\sqrt{L_t/2}}{N_t} \sqrt{d \sum_c n_{t,c}} = \sqrt{\frac{dL_t}{2N_t}}, \end{aligned}$$

where the second inequality is Cauchy–Schwarz. □

## 7 What distribution is being estimated?

The martingale theorem controls

$$\bar{p}_t(g) = \frac{1}{N_t} \sum_{k \in \mathcal{I}_t} p_k(g),$$

the running average of the origin-conditional bivariate CDFs over resolved origins $\mathcal{I}_t$. This distinction is fundamental.

### 7.1 Stable conditional CDF

**Assumption 5** (Stable conditional bivariate CDF)**.** *There exists a function $p : \mathcal{Q} \to [0,1]$ such that*

$$p_k(g) = p(g)$$

*for every origin $k$ and query $g \in \mathcal{Q}$.*

This is not an i.i.d. assumption. The realized pairs $B_k$ may be serially dependent and may overlap in physical time. The assumption says only that the origin-conditional lower-orthant probabilities are invariant.

**Corollary 6** (Delayed bivariate confidence band under conditional stability)**.** *Under Assumption 5, with probability at least $1-\delta$,*

$$\sup_{g \in \mathcal{Q}} |\widehat{p}_t(g) - p(g)| \le E_t \tag{18}$$

*simultaneously for all physical times $t$ with $N_t > 0$. Consequently,*

$$\underline{p}_t(g) := [\widehat{p}_t(g) - E_t]_+ , \tag{19}$$

$$\overline{p}_t(g) := [\widehat{p}_t(g) + E_t]_1 \tag{20}$$

*form simultaneous lower and upper confidence surfaces on $\mathcal{Q}$.*

Because the same scalar $E_t$ is subtracted from every grid point, the lower surface preserves coordinatewise monotonicity of the empirical bivariate CDF.

### 7.2 Changing conditional CDFs

If $p_k$ varies with $k$, the statistical theorem remains valid for the running average $\bar{p}_t$. To compare the empirical CDF with the current conditional CDF $p_t$, define the deterministic tracking discrepancy

$$B_t := \sup_{g \in \mathcal{Q}} |p_t(g) - \bar{p}_t(g)| . \tag{21}$$

Then on the same event,

$$\sup_{g \in \mathcal{Q}} |\widehat{p}_t(g) - p_t(g)| \le E_t + B_t. \tag{22}$$

Thus statistical martingale error and temporal tracking error separate additively.

A convenient sufficient condition is bounded one-step variation. Let

$$v_u := \sup_{g \in \mathcal{Q}} |p_{u+1}(g) - p_u(g)|. \tag{23}$$

Then

$$B_t \le \frac{1}{N_t} \sum_{k \in \mathcal{I}_t} \sum_{u=k}^{t-1} v_u. \tag{24}$$

This bound also shows why a full-history estimator may track a moving CDF poorly: old observations accumulate temporal bias.

## 8 Finite-horizon moving-window bound

For a drifting CDF, a recent-window estimator is more natural. This section gives a finite-horizon version with a sharp dependence on the window size. Fix a terminal physical horizon $T_{\max}$. Consider any consecutive interval of observations within a residue stream, of length $m$.

**Lemma 7** (Uniform concentration over residue intervals)**.** *Let*

$$C_0 := dQ.$$

*With probability at least $1-\delta$, every residue $c$, every query $g \in \mathcal{Q}$, and every consecutive interval of residue indices contained before $T_{\max}$ satisfy*

$$\left| \frac{1}{m} \sum_{i=a+1}^{a+m} D_{c,i}(g) \right| \le r_m^{\mathrm{win}}, \qquad r_m^{\mathrm{win}} := \sqrt{\frac{1}{2m} \log\left(\frac{C_0 T_{\max}^2}{\delta}\right)}. \tag{25}$$

*Proof.* For any fixed interval, conditional Hoeffding gives probability at most $2e^{-2m(r_m^{\mathrm{win}})^2} = 2\delta/(C_0 T_{\max}^2)$. There are at most $T_{\max}^2/2$ index intervals in each residue stream, and $C_0 = dQ$ residue-query pairs. A union bound gives total failure probability at most $\delta$. □

Now let $\widehat{p}_t^{(W)}(g)$ use the $W$ most recent fully resolved origins for the edge. Combining the residue streams as before yields the statistical bound

$$\sup_{g \in \mathcal{Q}} \left| \widehat{p}_t^{(W)}(g) - \bar{p}_t^{(W)}(g) \right| \le \sqrt{\frac{d}{2W} \log\left(\frac{dQT_{\max}^2}{\delta}\right)}. \tag{26}$$

Suppose additionally that

$$\sup_{g \in \mathcal{Q}} |p_{u+1}(g) - p_u(g)| \le \rho \qquad \text{for all } u. \tag{27}$$

The newest fully resolved origin is at least $d$ time steps old. The $W$ consecutive resolved origins have ages $d, d+1, \dots, d+W-1$, so

$$\sup_{g \in \mathcal{Q}} |p_t(g) - \bar{p}_t^{(W)}(g)| \le \rho \left( d + \frac{W-1}{2} \right). \tag{28}$$

Combining (26) and (28) gives the current-time bound

$$\boxed{\sup_{g \in \mathcal{Q}} \left| \widehat{p}_t^{(W)}(g) - p_t(g) \right| \le \sqrt{\frac{d}{2W} \log\left(\frac{dQT_{\max}^2}{\delta}\right)} + \rho \left( d + \frac{W-1}{2} \right).} \tag{29}$$

The first term is statistical uncertainty; the second is unavoidable tracking bias under the assumed drift model. Set

$$L_{\max} := \log\left(\frac{dQT_{\max}^2}{\delta}\right).$$

For the continuous relaxation in $W$, ignoring only the integer and feasibility constraints on the window length, the right-hand side is minimized at

$$W_\star = \left( \frac{dL_{\max}}{2\rho^2} \right)^{1/3}.$$

At this value, the optimized bound is

$$\rho\left(d-\frac{1}{2}\right)+\frac{3}{2}\left(\frac{\rho d L_{\max}}{2}\right)^{1/3}.$$

Thus a simpler order statement with the logarithmic factor shown explicitly is

$$O\left(\rho d+\left[\rho d\log\left(\frac{dQT_{\max}^2}{\delta}\right)\right]^{1/3}\right).$$

The additive $\rho d$ term is the irreducible staleness cost created by delayed revelation.

# 9 Adaptive aggregation over memory scales

The fixed-window bound above leaves a practical tuning question: which memory length $W$ should be used when the drift rate is unknown or changes over time? This section gives a rigorous aggregation construction. The statistical and online-learning guarantees are deliberately separated:

1. a single simultaneous martingale event controls *every* base window and therefore any data-dependent convex combination of them; and
2. a fixed-share multiplicative update chooses the mixture weights and enjoys interval-wise adaptive regret under delayed feedback.

The weight update is the fixed-share update used in the MARA meta-algorithm of Gradu, Hazan, and Minasyan [7]. We use only their multiplicative aggregation mechanism and interval-regret argument; the control-specific random expert-switching and action-shift machinery in that paper are not needed here because a convex combination of CDF estimates is itself a valid CDF estimate.

## 9.1 A dictionary of window experts

Fix a finite dictionary of window lengths

$$1\le W_1<\cdots<W_J,$$

and consider origins $k$ for which all $J$ windows are available, i.e. $k\ge d+W_J$. Expert $j$ uses the $W_j$ most recent fully resolved origins,

$$\mathcal{W}_{k,j}:=\{k-d-W_j+1,\dots,k-d\}, \tag{30}$$

and outputs the empirical bivariate CDF

$$\widehat{p}_{k,j}(g):=\frac{1}{W_j}\sum_{r\in\mathcal{W}_{k,j}}X_r(g),\qquad g\in\mathcal{Q}. \tag{31}$$

Its conditional-CDF target is

$$\bar{p}_{k,j}(g):=\frac{1}{W_j}\sum_{r\in\mathcal{W}_{k,j}}p_r(g). \tag{32}$$

Both $\widehat{p}_{k,j}$ and any function of previously resolved losses are $\mathcal{F}_k$-measurable. This predictability will be used below when the estimates are scored against the delayed observation generated at origin $k$.

Let $\pi_k = (\pi_{k,1}, \ldots, \pi_{k,J})$ be any random vector in the simplex $\Delta_J$ that is $\mathcal{F}_k$-measurable. Define the convex aggregate

$$\widehat{p}_k^{\mathrm{mix}}(g) := \sum_{j=1}^{J} \pi_{k,j} \widehat{p}_{k,j}(g), \tag{33}$$

$$\bar{p}_k^{\mathrm{mix}}(g) := \sum_{j=1}^{J} \pi_{k,j} \bar{p}_{k,j}(g). \tag{34}$$

Because each base estimate is coordinatewise nondecreasing in $g$, the mixture is also a bivariate CDF on the grid.

The mixture can equivalently be written as a learned weighting of observation ages. For $m = 0, \ldots, W_J - 1$, set

$$\omega_{k,m} := \sum_{j : W_j > m} \frac{\pi_{k,j}}{W_j}. \tag{35}$$

Then $\omega_{k,m} \geq 0$, $\sum_m \omega_{k,m} = 1$, and

$$\widehat{p}_k^{\mathrm{mix}}(g) = \sum_{m=0}^{W_J - 1} \omega_{k,m} X_{k-d-m}(g). \tag{36}$$

Moreover $\omega_{k,m+1} \leq \omega_{k,m}$: mixing uniform windows induces a nonincreasing recency profile automatically.

**Proposition 8** (Average age of the adaptive mixture)**.** *The induced weights satisfy*

$$A_k := \sum_{m=0}^{W_J - 1} \omega_{k,m}(d+m) = \sum_{j=1}^{J} \pi_{k,j} \left( d + \frac{W_j - 1}{2} \right). \tag{37}$$

*If the one-step CDF drift obeys*

$$\sup_{g \in \mathcal{Q}} |p_{u+1}(g) - p_u(g)| \leq \rho \qquad \text{for every } u,$$

*then*

$$\sup_{g \in \mathcal{Q}} |\bar{p}_k^{\mathrm{mix}}(g) - p_k(g)| \leq \rho A_k. \tag{38}$$

*Proof.* Substituting (35) and interchanging the finite sums,

$$\begin{aligned} \sum_m \omega_{k,m}(d+m) &= \sum_{j=1}^{J} \frac{\pi_{k,j}}{W_j} \sum_{m=0}^{W_j - 1} (d+m) \\ &= \sum_{j=1}^{J} \pi_{k,j} \left( d + \frac{W_j - 1}{2} \right). \end{aligned}$$

For every $m$ and $g$, the bounded-drift assumption and the triangle inequality give

$$|p_k(g) - p_{k-d-m}(g)| \leq \rho(d+m).$$

Using the weighted-history representation of the target and then averaging the last display with weights $\omega_{k,m}$ proves (38). □

### 9.2 Post-selection-safe confidence bands

The interval-uniform event from Lemma 7 already controls every consecutive window in every residue stream. Consequently, selecting or mixing window lengths after observing the past does not require another confidence split over the $J$ experts.

Set

$$L_{\max} := \log\left(\frac{dQT_{\max}^2}{\delta}\right), \qquad a_j := \sqrt{\frac{dL_{\max}}{2W_j}}. \tag{39}$$

**Theorem 9** (Adaptive convex aggregation on one simultaneous event)**.** *On the event of Lemma 7, simultaneously for every origin $k$ for which all base windows are defined, every $g \in \mathcal{Q}$, and every $\mathcal{F}_k$-measurable mixture vector $\pi_k \in \Delta_J$,*

$$\left|\widehat{p}_k^{\mathrm{mix}}(g) - \bar{p}_k^{\mathrm{mix}}(g)\right| \le A_k^{\mathrm{stat}}, \qquad A_k^{\mathrm{stat}} := \sum_{j=1}^{J} \pi_{k,j} a_j. \tag{40}$$

*Hence the statement holds with probability at least $1-\delta$ without an additional factor $J$ inside the confidence level.*

*If, in addition, the bounded-drift condition with parameter $\rho$ holds, then on the same event*

$$\boxed{\sup_{g\in\mathcal{Q}} \left|\widehat{p}_k^{\mathrm{mix}}(g) - p_k(g)\right| \le C_k^{\mathrm{mix}} := \sum_{j=1}^{J} \pi_{k,j} \left[\sqrt{\frac{dL_{\max}}{2W_j}} + \rho\left(d + \frac{W_j - 1}{2}\right)\right].} \tag{41}$$

*Consequently,*

$$\underline{p}_k^{\mathrm{mix}}(g) := \left[\widehat{p}_k^{\mathrm{mix}}(g) - C_k^{\mathrm{mix}}\right]_+, \tag{42}$$

$$\bar{p}_k^{\mathrm{mix}}(g) := \left[\widehat{p}_k^{\mathrm{mix}}(g) + C_k^{\mathrm{mix}}\right]_1 \tag{43}$$

*form simultaneous lower and upper confidence surfaces on $\mathcal{Q}$.*

*Proof.* For every base window $j$, the argument leading to (26) applies to that window on the single event of Lemma 7; therefore

$$\sup_{g\in\mathcal{Q}} |\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g)| \le a_j$$

simultaneously over all $k$ and $j$. The event is already uniform over all consecutive intervals, so introducing a finite dictionary of such windows does not create a new union bound.

For any realized $\pi_k \in \Delta_J$,

$$\begin{aligned}\left|\widehat{p}_k^{\mathrm{mix}}(g) - \bar{p}_k^{\mathrm{mix}}(g)\right| &\le \sum_{j=1}^{J} \pi_{k,j} |\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g)| \\ &\le \sum_{j=1}^{J} \pi_{k,j} a_j.\end{aligned}$$

This argument is pathwise on the simultaneous event, which is why the weights may be data dependent. Finally,

$$|\widehat{p}_k^{\mathrm{mix}}(g) - p_k(g)| \le |\widehat{p}_k^{\mathrm{mix}}(g) - \bar{p}_k^{\mathrm{mix}}(g)| + |\bar{p}_k^{\mathrm{mix}}(g) - p_k(g)|,$$

and Proposition 8 bounds the second term by $\rho A_k$. Substituting (37) proves (41). □

**Remark 10** (What adaptivity costs in the confidence radius)**.** *The confidence event itself pays no extra* $\log J$ *factor because Lemma 7 is uniform over all admissible intervals before the weights are chosen. The price of learning among* $J$ *memory scales appears instead in the adaptive-regret term below. The radius* $A_k^{\mathrm{stat}} = \sum_j \pi_{k,j} a_j$ *can be conservative compared with a concentration inequality tailored to one deterministic weighted average, but it remains valid under arbitrary data-dependent post-selection of the mixture weights.*

### 9.3 Fixed-share learning under delayed feedback

We next specify how to learn $\pi_k$. For each origin $k$ and expert $j$, store the prediction (31) made at origin $k$. When $B_k$ is revealed at physical time $k+d$, compute the integrated Brier loss

$$\ell_{k,j} := \frac{1}{Q} \sum_{g \in \mathcal{Q}} \left(X_k(g) - \widehat{p}_{k,j}(g)\right)^2 \in [0,1]. \tag{44}$$

For a mixture prediction, define analogously

$$\ell_k^{\mathrm{mix}} := \frac{1}{Q} \sum_{g \in \mathcal{Q}} \left(X_k(g) - \widehat{p}_k^{\mathrm{mix}}(g)\right)^2. \tag{45}$$

A single global expert learner would receive these losses with delay $d$. Instead, use the same residue decomposition that produced the martingale filtration. For residue $c \in \{1, \dots, d\}$, write its eligible origins as

$$k_{c,1} < k_{c,2} < \cdots, \qquad k_{c,i+1} = k_{c,i} + d.$$

The outcome from $k_{c,i}$ is available at $k_{c,i} + d = k_{c,i+1}$, before the next prediction in that residue stream. Thus each residue stream is a standard full-information expert problem in its own meta-time index.

Let

$$n_{\max} := \left\lceil \frac{T_{\max}}{d} \right\rceil, \qquad \sigma := \frac{1}{n_{\max}}, \qquad 0 < \eta \le \frac{1}{2},$$

and assume $n_{\max} \ge 2$. For every residue, initialize $w_{c,1,j} = 1$. At meta-round $i$, use

$$\pi_{k_{c,i},j} = \frac{w_{c,i,j}}{\sum_{\ell=1}^{J} w_{c,i,\ell}}, \tag{46}$$

form the deterministic mixture (33), and, when the loss vector from that origin is revealed, update

$$\widetilde{w}_{c,i+1,j} := w_{c,i,j} \exp(-\eta \ell_{k_{c,i},j}), \tag{47}$$

$$w_{c,i+1,j} := (1-\sigma)\widetilde{w}_{c,i+1,j} + \frac{\sigma}{J} \sum_{\ell=1}^{J} \widetilde{w}_{c,i+1,\ell}. \tag{48}$$

This is the fixed-share multiplicative update used in Algorithm 2 of Gradu, Hazan, and Minasyan [7]. Their random expert-selection step is unnecessary here because we can output the convex mixture directly.

**Theorem 11** (Residue-wise adaptive regret for the CDF mixture)**.** *Fix a residue* $c$ *and any contiguous meta-time interval* $I_c = [r, s]$*. Under updates (47)–(48),*

$$\boxed{(1-\eta) \sum_{i=r}^{s} \ell_{k_{c,i}}^{\mathrm{mix}} \le \min_{j \in [J]} \sum_{i=r}^{s} \ell_{k_{c,i},j} + \frac{\log(n_{\max} J) + 2}{\eta}.} \tag{49}$$

*Thus the learned mixture competes, on every local interval of a residue stream, with the best fixed memory scale on that interval.*

*Proof.* This is a direct specialization of the fixed-share potential argument in Lemma A.2 of Gradu, Hazan, and Minasyan [7]; we reproduce the steps to make the constants and the mapping to the CDF problem explicit.

Suppress the residue index and write $W_i := \sum_j w_{i,j}$ and $\pi_{i,j} := w_{i,j}/W_i$. The sharing step preserves total weight, so

$$\frac{W_{i+1}}{W_i} = \sum_{j=1}^{J} \pi_{i,j} e^{-\eta \ell_{i,j}}. \tag{50}$$

Because $0 \le \ell_{i,j} \le 1$ and $0 < \eta \le 1/2$, $e^{-x} \le 1 - x + x^2$ for $x \in [0,1]$, and $\ell_{i,j}^2 \le \ell_{i,j}$. Hence, with $m_i := \sum_j \pi_{i,j} \ell_{i,j}$,

$$\begin{aligned} \frac{W_{i+1}}{W_i} &\le 1 - \eta m_i + \eta^2 m_i \\ &\le \exp\{-\eta(1-\eta) m_i\}. \end{aligned}$$

Multiplying over $i = r, \dots, s$ gives

$$\log \frac{W_{s+1}}{W_r} \le -\eta(1-\eta) \sum_{i=r}^{s} m_i. \tag{51}$$

For any fixed expert $j$, the sharing update implies

$$w_{i+1,j} \ge (1-\sigma) w_{i,j} e^{-\eta \ell_{i,j}}.$$

Iterating over $I_c$ and using $W_{s+1} \ge w_{s+1,j}$ yields

$$\log \frac{W_{s+1}}{W_r} \ge \log \pi_{r,j} + |I_c| \log(1-\sigma) - \eta \sum_{i=r}^{s} \ell_{i,j}. \tag{52}$$

After every share step, $\pi_{i,j} \ge \sigma/J$, while the initialization is uniform, so $\pi_{r,j} \ge \sigma/J$. With $\sigma = 1/n_{\max}$, $|I_c| \le n_{\max}$, and $\log(1-\sigma) \ge -2\sigma$ for $\sigma \le 1/2$, the right-hand side of (52) is at least

$$-\log(n_{\max} J) - 2 - \eta \sum_{i=r}^{s} \ell_{i,j}.$$

Combining with (51), dividing by $\eta$, and minimizing over $j$ gives

$$(1-\eta) \sum_{i=r}^{s} m_i \le \min_j \sum_{i=r}^{s} \ell_{i,j} + \frac{\log(n_{\max} J) + 2}{\eta}. \tag{53}$$

Finally, squared loss is convex in the prediction. Therefore Jensen's inequality gives, at every origin,

$$\ell_{k_{c,i}}^{\mathrm{mix}} \le \sum_j \pi_{i,j} \ell_{i,j} = m_i.$$

Substitution into (53) proves the theorem. □

**Corollary 12** (Adaptive regret on a physical-time interval)**.** *Let $I = [r, s]$ be any physical-time interval after the common burn-in and let $d_I$ denote the number of residue classes represented in $I$; hence $d_I \le \min\{d, |I|\}$. Then*

$$\boxed{(1-\eta)\sum_{k\in I}\ell_k^{\mathrm{mix}} \le \min_{j\in[J]}\sum_{k\in I}\ell_{k,j} + \frac{d_I\,[\log(n_{\max}J)+2]}{\eta}.} \tag{54}$$

*Proof.* The intersection of a physical interval with any residue class is a contiguous meta-time interval. Apply Theorem 11 to every nonempty residue intersection and sum. This gives the sum, over residues, of their individual best-expert losses plus $d_I$ copies of the complexity term. For every common expert $j$, the sum of the residue-wise minima is no larger than that expert's loss over all of $I$; minimizing over the common $j$ gives (54). □

### 9.4 From Brier adaptive regret to CDF estimation error

The Brier loss has a direct statistical interpretation in this problem. For any $\mathcal{F}_k$-measurable CDF prediction $q : \mathcal{Q} \to [0, 1]$, define its grid-averaged squared estimation error

$$\mathcal{R}_k(q) := \frac{1}{Q}\sum_{g\in\mathcal{Q}}(q(g) - p_k(g))^2. \tag{55}$$

Also define the irreducible Bernoulli variance term

$$\nu_k := \frac{1}{Q}\sum_{g\in\mathcal{Q}} p_k(g)(1-p_k(g)) \le \frac{1}{4}. \tag{56}$$

**Remark 13** (Interpretation as excess risk relative to the true CDF)**.** *The risk*

$$\mathcal{R}_k(q) = \frac{1}{Q}\sum_{g\in\mathcal{Q}}\big(q(g) - p_k(g)\big)^2$$

*has a natural clairvoyant benchmark. An oracle that knows the current conditional CDF $p_k$ exactly would choose $q = p_k$ and incur zero risk:*

$$\mathcal{R}_k(p_k) = 0.$$

*Consequently,*

$$\sum_{k\in I}\mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\mathrm{mix}})\big]$$

*is the cumulative expected excess CDF-estimation risk relative to a clairvoyant oracle that knows $p_k$ at every origin.*

*The oracle inequalities developed below therefore admit two complementary interpretations. First, the adaptive mixture competes with the best fixed memory scale in the chosen window dictionary. Second, after bounding the risk of that best window, the resulting bound directly controls the aggregate's absolute gap to the zero-risk truth oracle. In this decomposition, the best-window term captures the intrinsic statistical and tracking difficulty of estimating the time-varying CDF, while the remaining meta-learning terms quantify the additional price of adaptively learning an appropriate memory scale.*

**Lemma 14** (Conditional Brier decomposition)**.** *For every $\mathcal{F}_k$-measurable prediction $q$,*

$$\mathbb{E}\left[\frac{1}{Q}\sum_{g\in\mathcal{Q}}(X_k(g)-q(g))^2 \,\middle|\, \mathcal{F}_k\right] = \nu_k + \mathcal{R}_k(q). \tag{57}$$

*Proof.* For a fixed $g$, $X_k(g)\in\{0,1\}$ and $\mathbb{E}[X_k(g)\mid\mathcal{F}_k]=p_k(g)$. Expanding around the conditional mean,

$$\begin{aligned}\mathbb{E}[(X_k(g)-q(g))^2\mid\mathcal{F}_k] &= \mathrm{Var}(X_k(g)\mid\mathcal{F}_k)+(p_k(g)-q(g))^2\\ &= p_k(g)(1-p_k(g))+(p_k(g)-q(g))^2.\end{aligned}$$

Average over $g\in\mathcal{Q}$. □

**Theorem 15** (Local oracle inequality for adaptive CDF estimation)**.** *For every physical interval $I=[r,s]$ after burn-in, the residue-wise fixed-share aggregate satisfies*

$$\boxed{\sum_{k\in I}\mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\mathrm{mix}})\big] \le \frac{1}{1-\eta}\left\{\min_{j\in[J]}\sum_{k\in I}\mathbb{E}[\mathcal{R}_k(\widehat{p}_{k,j})]+\frac{d_I\,[\log(n_{\max}J)+2]}{\eta}+\frac{\eta|I|}{4}\right\}.} \tag{58}$$

*Thus, on every local interval, the aggregate tracks the best memory scale in expected grid-averaged squared CDF error, up to an explicit adaptive-regret penalty. No drift parameter $\rho$ is required to run the weighting scheme.*

*Proof.* Corollary 12 is a pathwise inequality in the realized Brier losses. Take expectations of both sides. Since $\mathbb{E}[\min_j Y_j]\le\min_j\mathbb{E}[Y_j]$,

$$(1-\eta)\sum_{k\in I}\mathbb{E}[\ell_k^{\mathrm{mix}}] \le \min_j\sum_{k\in I}\mathbb{E}[\ell_{k,j}]+\frac{d_I[\log(n_{\max}J)+2]}{\eta}.$$

Every base prediction and the mixture prediction are $\mathcal{F}_k$-measurable by construction. Applying Lemma 14 to each term gives

$$(1-\eta)\sum_{k\in I}\mathbb{E}[\nu_k+\mathcal{R}_k(\widehat{p}_k^{\mathrm{mix}})] \le \sum_{k\in I}\mathbb{E}[\nu_k]+\min_j\sum_{k\in I}\mathbb{E}[\mathcal{R}_k(\widehat{p}_{k,j})]+\frac{d_I[\log(n_{\max}J)+2]}{\eta}.$$

Move the common variance contribution on the left to the right. The remainder is $\eta\sum_{k\in I}\mathbb{E}[\nu_k]\le\eta|I|/4$ by (56). Dividing by $1-\eta$ proves (58). □

### 9.5 Variation-budget view of nonstationarity

The pointwise drift envelope $v_u\le\rho$ is convenient for a certified current-CDF band, but it can be unnecessarily restrictive for regret analysis: it rules out a rare large jump even when the process is otherwise stable. A more flexible description is the cumulative path variation

$$v_u := \sup_{g\in\mathcal{Q}}|p_{u+1}(g)-p_u(g)|, \qquad V_{a:b}:=\sum_{u=a}^{b-1}v_u, \quad a<b, \tag{59}$$

with $V_{a:b}:=0$ when $b\le a$. This is a lower-orthant CDF variation budget, not total variation distance between the full laws. The use of a cumulative variation budget to quantify nonstationarity

is standard in online and stochastic optimization; see, for example, Besbes, Gur, and Zeevi [10]. Their variation metric and optimization problem are different from ours, so none of their minimax regret rates are invoked below. Here the variation budget is used only to control the tracking error of a delayed CDF estimator.

The bounded-drift model is the special case

$$V_{a:b} \le \rho(b-a).$$

The cumulative formulation also permits abrupt changes followed by long stable periods.

**Proposition 16** (Tracking error under an arbitrary variation path)**.** *For the adaptive mixture in* (36)*, define*

$$B_k^{\mathrm{mix}} := \sup_{g\in\mathcal{Q}} |\bar{p}_k^{\mathrm{mix}}(g) - p_k(g)|.$$

*Then, pathwise,*

$$\boxed{B_k^{\mathrm{mix}} \le \sum_{m=0}^{W_J-1} \omega_{k,m} V_{k-d-m:k}.} \tag{60}$$

*For a single window expert $j$,*

$$B_{k,j} := \sup_{g\in\mathcal{Q}} |\bar{p}_{k,j}(g) - p_k(g)| \le \frac{1}{W_j} \sum_{m=0}^{W_j-1} V_{k-d-m:k}. \tag{61}$$

*Consequently, if $v_u \le \rho$ for every $u$, then* (60) *reduces to $B_k^{\mathrm{mix}} \le \rho A_k$, recovering Proposition 8.*

*Proof.* For every $m$ and $g$, repeated triangle inequality gives

$$|p_k(g) - p_{k-d-m}(g)| \le \sum_{u=k-d-m}^{k-1} \sup_{h\in\mathcal{Q}} |p_{u+1}(h) - p_u(h)| = V_{k-d-m:k}.$$

The target mixture has the same age weights as (36), namely $\bar{p}_k^{\mathrm{mix}}(g) = \sum_m \omega_{k,m} p_{k-d-m}(g)$. Taking the weighted average of the preceding inequality proves (60). Uniform weights $\omega_{k,m} = 1/W_j$ for $m = 0, \dots, W_j - 1$ give (61). Finally, $V_{k-d-m:k} \le \rho(d+m)$ under $v_u \le \rho$; averaging over $m$ gives $\rho A_k$ by (37). □

The next lemma quantifies the stochastic part of a fixed-window estimator in mean square. It uses only the staggered martingale-difference construction; no mixing assumption is needed.

**Lemma 17** (Mean-square fluctuation of one window expert)**.** *For every base window $j$ and eligible origin $k$,*

$$\mathbb{E}\left[\frac{1}{Q}\sum_{g\in\mathcal{Q}} \left(\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g)\right)^2\right] \le \frac{d}{4W_j}. \tag{62}$$

*Proof.* Fix $g \in \mathcal{Q}$ and partition the $W_j$ origins in $\mathcal{W}_{k,j}$ by their residue modulo $d$. For residue $c$, let

$$S_c(g) := \sum_{r\in\mathcal{W}_{k,j}:\, r\equiv c \pmod d} \big(X_r(g) - p_r(g)\big).$$

Then

$$\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g) = \frac{1}{W_j} \sum_{c=1}^{d} S_c(g).$$

By Cauchy–Schwarz, $(\sum_c S_c)^2 \le d \sum_c S_c^2$. Within each residue stream, the summands are martingale differences with respect to the staggered filtration. Distinct increments in the same stream are therefore orthogonal in $L^2$, and for a Bernoulli increment

$$\mathbb{E}[(X_r(g) - p_r(g))^2] = \mathbb{E}[p_r(g)(1 - p_r(g))] \le \frac{1}{4}.$$

If $n_c$ window observations lie in residue $c$, then $\mathbb{E}[S_c(g)^2] \le n_c/4$. Since $\sum_c n_c = W_j$,

$$\mathbb{E}\big[(\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g))^2\big] \le \frac{d}{4W_j}.$$

Averaging over $g \in \mathcal{Q}$ proves the lemma. □

**Lemma 18** (Cumulative tracking cost of a fixed window)**.** *Let $I = [r, s]$ be a physical-time interval after burn-in and set $n_I := |I|$. For expert $j$, define the enlarged-interval variation*

$$\mathcal{V}_{I,j} := V_{r-d-W_j+1:s} = \sum_{u=r-d-W_j+1}^{s-1} v_u, \tag{63}$$

*and its average age*

$$A_j := d + \frac{W_j - 1}{2}.$$

*Then, pathwise,*

$$\boxed{\sum_{k \in I} B_{k,j} \le A_j \, \mathcal{V}_{I,j}.} \tag{64}$$

*Proof.* By (61),

$$\sum_{k=r}^{s} B_{k,j} \le \frac{1}{W_j} \sum_{m=0}^{W_j-1} \sum_{k=r}^{s} \sum_{u=k-d-m}^{k-1} v_u.$$

Fix $m$. A particular increment $v_u$ can appear in the inner sum only for $k \in \{u+1, \dots, u+d+m\}$, hence at most $d+m$ times. All such increments lie inside $[r-d-W_j+1, s-1]$. Therefore

$$\sum_{k=r}^{s} \sum_{u=k-d-m}^{k-1} v_u \le (d+m) \, \mathcal{V}_{I,j}.$$

Averaging over $m = 0, \dots, W_j - 1$ yields

$$\sum_{k \in I} B_{k,j} \le \left[ \frac{1}{W_j} \sum_{m=0}^{W_j-1} (d+m) \right] \mathcal{V}_{I,j} = A_j \mathcal{V}_{I,j}.$$

□

**Theorem 19** (Variation-budget oracle bound for the adaptive estimator)**.** *Let $I = [r, s]$ be any physical interval after burn-in, with $n_I = |I|$. For each expert $j$, let $\mathcal{V}_{I,j}$ be defined by (63). Then the residue-wise fixed-share aggregate satisfies*

$$\boxed{\sum_{k\in I} \mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\mathrm{mix}})\big] \le \frac{1}{1-\eta}\left[\min_{j\in[J]}\left\{\frac{n_I d}{2W_j} + 2A_j\,\mathbb{E}[\mathcal{V}_{I,j}]\right\} + \frac{d_I[\log(n_{\max}J)+2]}{\eta} + \frac{\eta n_I}{4}\right].} \tag{65}$$

*If a deterministic pathwise budget $\mathcal{V}_{I,j} \le V_I$ holds for every candidate window, then $\mathbb{E}[\mathcal{V}_{I,j}]$ may be replaced by $V_I$. The meta-learner does not need to know $V_I$.*

*Proof.* For any $g$,

$$\widehat{p}_{k,j}(g) - p_k(g) = \big(\widehat{p}_{k,j}(g) - \bar{p}_{k,j}(g)\big) + \big(\bar{p}_{k,j}(g) - p_k(g)\big).$$

Using $(x+y)^2 \le 2x^2 + 2y^2$, averaging over $g$, taking expectations, and applying Lemma 17 gives

$$\mathbb{E}[\mathcal{R}_k(\widehat{p}_{k,j})] \le \frac{d}{2W_j} + 2\mathbb{E}[B_{k,j}^2]. \tag{66}$$

Because two CDF values differ by at most one, $0 \le B_{k,j} \le 1$, so $B_{k,j}^2 \le B_{k,j}$. Summing (66) over $k \in I$ and using Lemma 18 yields

$$\sum_{k\in I} \mathbb{E}[\mathcal{R}_k(\widehat{p}_{k,j})] \le \frac{n_I d}{2W_j} + 2A_j\mathbb{E}[\mathcal{V}_{I,j}]. \tag{67}$$

Substitute (67) into the local oracle inequality (58). The latter rests on the residue-wise fixed-share adaptive-regret argument of Gradu, Hazan, and Minasyan [7], specialized and proved above. This proves (65). The deterministic-budget statement is immediate. □

**Corollary 20** (Rate suggested by a common variation budget)**.** *Suppose $\mathcal{V}_{I,j} \le V_I$ for every candidate window and that the window dictionary contains a value within a constant factor of*

$$W_I^* := \sqrt{\frac{n_I d}{2V_I}},$$

*whenever $0 < V_I < \infty$ and this value lies in the admissible window range. Then the best-window term in (65) is, up to a universal constant from discretizing the window dictionary,*

$$O\Big(\sqrt{n_I d V_I} + dV_I\Big). \tag{68}$$

*For a geometric dictionary $W_j = 2^{j-1}$, the constant-factor coverage of $W_I^*$ is automatic away from the endpoints. Thus the meta-learner adapts to the unknown variation scale through its competition with the best memory length; no value of $V_I$ or $\rho$ is required by the update itself.*

*Proof.* Under the common budget, the window-dependent part of (65) is

$$\frac{n_I d}{2W} + 2\left(d + \frac{W-1}{2}\right)V_I = \frac{n_I d}{2W} + WV_I + (2d-1)V_I.$$

Ignoring the additive term independent of $W$, the continuous minimizer of $\frac{n_I d}{2W} + WV_I$ is $W_I^* = \sqrt{n_I d/(2V_I)}$, and the two $W$-dependent terms sum to $\sqrt{2n_I d V_I}$. Choosing a dictionary element within a constant factor of $W_I^*$ changes this quantity by only a universal constant. Adding the $(2d-1)V_I$ term proves (68). □

**Remark 21** (When sublinear estimation error is possible)**.** *For fixed delay $d$, the variation-dependent contribution in (68) is sublinear in $n_I$ whenever $V_I = o(n_I)$. This parallels the general role of sublinear variation budgets in nonstationary regret theory, but should not be read as an application of the minimax results of Besbes, Gur, and Zeevi [10]: our target, feedback structure, loss, and variation metric are different. If the variation budget is linear in the interval length, the bound permits linear cumulative estimation error, which is the expected qualitative behavior in a persistently changing environment.*

**Remark 22** (Interpretation of the two guarantees)**.** *Theorem 9 and Theorem 15 answer different questions. The first is a high-probability $\ell_\infty(\mathcal{Q})$ confidence statement and still requires a temporal regularity condition, such as a known $\rho$, if the target is the current CDF $p_k$. The second is an adaptive prediction guarantee: the fixed-share weights need not know $\rho$ and compete locally with the best window expert in expected squared CDF error. Thus one may use the same adaptive mixture either in a certified mode, when a valid drift envelope is available, or as a plug-in estimator whose tracking quality is controlled by adaptive regret.*

**Remark 23** (Choice of the memory dictionary)**.** *A convenient implementation uses geometric windows, for example $W_j = 2^{j-1}$ up to the largest feasible window. Then $J = O(\log T_{\max})$, so the expert-selection complexity enters the regret bound only through $\log J$. Under the bounded-drift model, writing $L_{\max} = \log(dQT_{\max}^2/\delta)$, the continuous-relaxation optimizer of the fixed-window confidence radius is $W_\star = (dL_{\max}/(2\rho^2))^{1/3}$ as derived above. A geometric dictionary contains a window within a constant factor of that scale without requiring the algorithm to know $\rho$ in advance; Theorem 15 explains how the Brier-weighted aggregate can track whichever available scale predicts best on each local interval.*

## 10 Simultaneous control of many edges

For applications with a set $\mathcal{E}$ of pairwise processes, let edge $e$ have delay $d_e$ and query set $\mathcal{Q}_e$ of size $Q_e$. Define the global complexity count

$$C := \sum_{e \in \mathcal{E}} d_e Q_e. \tag{69}$$

The proof of Theorem 2 gives the following result without requiring independence across edges.

**Corollary 24** (One global confidence event over edges)**.** *For*

$$r_n^{\mathrm{glob}}(\delta) := \sqrt{\frac{1}{2n} \log\left(\frac{\pi^2 C n^2}{3\delta}\right)}, \tag{70}$$

*with probability at least $1 - \delta$, the residue-level bounds hold simultaneously for every edge $e$, every one of its $d_e$ residues, every query $g \in \mathcal{Q}_e$, and every sample size $n \geq 1$.*

*Consequently, if $N_{t,e}$ edge-$e$ observations have resolved by time $t$, then*

$$\boxed{\sup_{g \in \mathcal{Q}_e} |\widehat{p}_{t,e}(g) - \bar{p}_{t,e}(g)| \leq \sqrt{\frac{d_e}{2N_{t,e}} \log\left(\frac{\pi^2 C N_{t,e}^2}{3\delta}\right)}} \tag{71}$$

*simultaneously over all edges and times.*

This form uses one global confidence level $\delta$. The multiplicity over edges, delays, and bivariate grid points enters only through the logarithm of $C$.

## 11 Implementation

The theorem is compatible with a simple streaming implementation. For each edge $e$:

1. maintain an $M \times M$ histogram of resolved pairs $B_k$;
2. increment one histogram cell whenever a new pair resolves;
3. obtain the empirical bivariate CDF on the grid by a two-dimensional cumulative sum;
4. maintain only the residue sample counts $n_{t,e,c}$, not separate histograms for each residue;
5. compute the exact statistical width

$$E_{t,e} = \sum_c \frac{n_{t,e,c}}{N_{t,e}} r^{\mathrm{glob}}_{n_{t,e,c}}(\delta),$$

   or the simpler upper bound in (71);
6. return either the plug-in surface $\widehat{p}_{t,e}$ or the one-sided lower surface

$$\underline{p}_{t,e}(g) = [\widehat{p}_{t,e}(g) - E_{t,e}]_+.$$

Because every resolved bivariate observation determines all lower-orthant indicators simultaneously, a single pair updates the entire empirical CDF. The grid size affects storage and numerical resolution, while its statistical cost is only logarithmic through $Q = M^2$.

With a dense graph of pairwise processes, storage is $O(|\mathcal{E}|M^2)$ for the histograms plus $O(\sum_e d_e)$ residue counts. Both the histogram updates and cumulative sums are amenable to batched tensor/GPU implementations.

## 12 Why the finite grid is theoretically useful

In the i.i.d. setting, bivariate empirical CDFs admit continuum-uniform Glivenko–Cantelli/DKW-type theory. Under arbitrary adapted dependence, the correct complexity notion changes. Rakhlin et al. show that sequential uniform convergence is characterized by sequential complexity measures, and explicitly note that even the ordinary one-dimensional threshold class can fail unrestricted sequential uniform convergence [1].

This creates a genuine obstacle to a completely distribution-free statement of the form

$$\sup_{a,b\in\mathbb{R}} |\widehat{p}_t(a,b) - \bar{p}_t(a,b)| \le \epsilon_t$$

under arbitrary adaptation. Since bivariate lower orthants contain one-dimensional threshold subclasses, a continuum extension needs additional structure or a more specialized argument.

The finite grid removes this ambiguity: it creates a finite function class for which the martingale concentration argument is immediate and fully explicit. For online allocation applications this is often sufficient because the same grid is already used numerically. A continuum extension under restricted adversaries, mixing, stability, or instance-dependent sequential complexity is a separate theoretical direction.

## 13 Discussion and open questions

The note establishes a conservative but transparent delayed bivariate martingale CDF theory. Several improvements are possible.

1. **Sharper treatment of delay.** The $\sqrt{d/t}$ rate is produced by the staggered proof. A specialized delayed-martingale argument may combine all origins more efficiently and reduce the delay penalty. The present theorem should therefore be viewed as a rigorous baseline, not a minimax statement.

2. **Continuum-uniform lower orthants.** It is worth identifying assumptions under which the bivariate lower-orthant class has tractable sequential complexity. Such a result could remove the explicit grid from the theorem.

3. **Time-uniform moving windows.** The moving-window result above is uniform over windows only up to a fixed terminal horizon $T_{\max}$. Confidence-sequence stitching could replace this with a fully anytime windowed result.

4. **Data-dependent widths.** Empirical-Bernstein or variance-adaptive martingale bounds may be much tighter when joint failures are rare, as is typical in low-miscoverage applications.

5. **Sharper adaptive aggregation.** Section 9 uses residue-wise fixed-share aggregation because it converts the physical delay into ordinary full-information meta-rounds. A delayed-experts analysis that shares one global weight vector across residues could potentially reduce the explicit $d$ factor in the adaptive-regret term.

6. **Plug-in versus certified use.** In applications with a separate outer calibration controller, the empirical CDF may be used directly for efficiency while the lower confidence surface is retained as an optional conservative mode. The same bound quantifies the estimation error in either case.

## 14 Summary

For delayed bivariate online observations, the key construction is

$$\text{delay staggering} \implies \text{martingale-difference streams} \implies \text{uniform finite-grid CDF bound.}$$

The main edge-level result is

$$\sup_{g\in\mathcal{Q}} |\widehat{p}_t(g) - \bar{p}_t(g)| \le \sqrt{\frac{d}{2N_t}\log\left(\frac{\pi^2 dQN_t^2}{3\delta}\right)}$$

with one global confidence event. When $N_t \asymp t$, this is

$$O\left(\sqrt{\frac{d}{t}\log\left(\frac{dQt^2}{\delta}\right)}\right).$$

Under a stable conditional CDF, this is a confidence band for the common bivariate CDF. Under drift, the same statistical term combines with an explicit tracking-bias term. A dictionary of moving-window CDF estimators can additionally be combined with data-dependent fixed-share weights: the simultaneous confidence event survives post-selection, while the residue-wise meta-learner competes

on every interval with the best memory scale in Brier loss and in expected grid-averaged squared CDF error. Under a cumulative lower-orthant variation budget $V_I$, this yields a best-window contribution of order $\sqrt{|I|dV_I} + dV_I$ up to constants and the separate meta-regret term. The construction requires no i.i.d. assumption and treats delay as a filtration problem rather than as an independence assumption.

## 15 Summary of regret guarantees and assumptions

This section collects the regret guarantees in one place and separates the assumptions required for the basic adaptive-regret statement from those needed only for the variation-budget rate.

Let $I = [r, s]$ be a physical-time interval after burn-in, $n_I := |I|$, and let $d_I$ denote the number of residue classes represented in $I$. Recall the current-CDF estimation risk

$$\mathcal{R}_k(q) := \frac{1}{Q} \sum_{g \in \mathcal{Q}} \big(q(g) - p_k(g)\big)^2.$$

Since $\mathcal{R}_k(p_k) = 0$, the quantity

$$\sum_{k \in I} \mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\,\mathrm{mix}})\big]$$

is also the cumulative expected excess CDF-estimation risk relative to a clairvoyant oracle that knows the current conditional CDF $p_k$ at every origin.

**1. Pathwise adaptive regret in Brier loss.** For the residue-wise fixed-share mixture,

$$\boxed{(1-\eta) \sum_{k \in I} \ell_k^{\mathrm{mix}} \le \min_{j \in [J]} \sum_{k \in I} \ell_{k,j} + \frac{d_I[\log(n_{\max} J) + 2]}{\eta}.}$$

Thus, on every physical-time interval, the adaptive mixture competes with the best fixed memory scale in the dictionary on that interval. This is an adaptive-regret guarantee in the sense of local comparison on arbitrary contiguous intervals; such interval-wise regret criteria are standard in adaptive online learning [5, 6, 7].

**2. Oracle inequality for current-CDF estimation.** Using the conditional Brier decomposition gives

$$\boxed{\sum_{k \in I} \mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\,\mathrm{mix}})\big] \le \frac{1}{1-\eta} \left[ \min_{j \in [J]} \sum_{k \in I} \mathbb{E}[\mathcal{R}_k(\widehat{p}_{k,j})] + \frac{d_I[\log(n_{\max} J) + 2]}{\eta} + \frac{\eta n_I}{4} \right].}$$

The left-hand side measures distance to the zero-risk truth oracle $p_k$. The first term on the right is the estimation error of the best available window in hindsight, while the remaining terms are the additional cost of adaptively learning the appropriate memory scale.

**3. Variation-budget bound.** Let

$$v_u := \sup_{g \in \mathcal{Q}} |p_{u+1}(g) - p_u(g)|$$

and

$$\mathcal{V}_{I,j} := V_{r-d-W_j+1:s}.$$

With

$$A_j := d + \frac{W_j - 1}{2},$$

the adaptive estimator satisfies

$$\boxed{\sum_{k \in I} \mathbb{E}\big[\mathcal{R}_k(\widehat{p}_k^{\mathrm{mix}})\big] \le \frac{1}{1-\eta}\left[\min_{j \in [J]} \left\{ \frac{n_I d}{2W_j} + 2A_j \mathbb{E}[\mathcal{V}_{I,j}] \right\} + \frac{d_I[\log(n_{\max} J) + 2]}{\eta} + \frac{\eta n_I}{4}\right].}$$

The term $n_I d/(2W_j)$ is the statistical fluctuation cost of window $j$, while $2A_j\mathbb{E}[\mathcal{V}_{I,j}]$ is its tracking cost under temporal nonstationarity.

If a common pathwise budget

$$\mathcal{V}_{I,j} \le V_I \qquad \text{for all } j$$

is available, the window-dependent part is

$$\frac{n_I d}{2W_j} + 2\left(d + \frac{W_j - 1}{2}\right) V_I.$$

Its continuous optimal scale is

$$W_I^\star = \sqrt{\frac{n_I d}{2V_I}},$$

and a dictionary containing a window within a constant factor of $W_I^\star$ yields

$$\boxed{O\Big(\sqrt{n_I d V_I} + d V_I\Big)}$$

for the best-window contribution, in addition to the fixed-share meta-regret terms. Variation budgets are a standard way of describing nonstationarity, although the lower-orthant CDF variation used here differs from the cost-function variation studied, for example, by Besbes, Gur, and Zeevi [10].

### Assumptions used by the regret analysis

1. **Adapted observations and conditional-CDF target.** For every query $g \in \mathcal{Q}$,

   $$p_k(g) = \mathbb{E}[X_k(g) \mid \mathcal{F}_k].$$

   The process may otherwise depend arbitrarily on the past. No i.i.d., mixing, or independence-across-time assumption is imposed. Martingale formulations based on conditional distributions under adapted dependence are also used in [1, 2, 3].

2. **Deterministic bounded reveal delay.** Each observation used by the edge is completely revealed after delay $d$. The residue construction converts this delayed problem into $d$ ordinary full-information expert streams. Online learning with delayed feedback and reductions from delayed to non-delayed learning are studied more generally by Joulani, Gyorgy, and Szepesvari [8]. For a pairwise edge $e = (h_1, h_2)$ in the trajectory application, the natural choice is

   $$d_e = \max\{d_{h_1}, d_{h_2}\}.$$

3. **Finite query grid.** The regret loss is evaluated on a finite set $\mathcal{Q}$ of size $Q$:

$$\ell_{k,j} = \frac{1}{Q} \sum_{g \in \mathcal{Q}} (X_k(g) - \widehat{p}_{k,j}(g))^2.$$

This is both the computational representation of the bivariate CDF and a way of avoiding additional sequential-complexity assumptions required for unrestricted continuum-uniform empirical-process results under arbitrary dependence; see [1]. The regret results therefore make no claim about a continuum of thresholds outside $\mathcal{Q}$.

4. **Finite dictionary of memory scales.** The learner competes with a finite collection

$$1 \le W_1 < \cdots < W_J.$$

Finite expert classes and multiplicative-weight aggregation are standard in prediction with expert advice [4]. The fixed-share construction used to track locally best experts originates with Herbster and Warmuth [5]; our particular interval-regret argument follows the fixed-share construction used by Gradu, Hazan, and Minasyan [7].

5. **Predictable mixture weights.** The mixture vector $\pi_k$ is $\mathcal{F}_k$-measurable: it may depend on all information available before the outcome generated at origin $k$ is revealed, but not on that unrevealed outcome. This is the standard non-anticipation condition in online prediction with expert advice [4].

6. **Bounded Brier loss.** Every base prediction and mixture prediction takes values in $[0, 1]$, giving

$$0 \le \ell_{k,j} \le 1.$$

This boundedness is used in the multiplicative-weight potential argument. The Brier score is a classical proper scoring rule for probabilistic predictions; see Gneiting and Raftery [9]. Its conditional decomposition is especially convenient here because the excess expected Brier loss relative to the true conditional probability is exactly squared estimation error.

7. **Fixed-share tuning and finite meta-horizon.** The displayed constants use

$$n_{\max} = \left\lceil \frac{T_{\max}}{d} \right\rceil, \qquad \sigma = \frac{1}{n_{\max}}, \qquad 0 < \eta \le \frac{1}{2}.$$

These are algorithmic tuning conditions rather than distributional assumptions. Fixed-share and related expert-tracking methods use analogous sharing parameters [5, 7].

8. **Variation is required only for the explicit truth-oracle rate.** The fixed-share adaptive-regret and local oracle inequalities do *not* assume a drift bound, $\rho$, or a variation budget. The quantities $\mathcal{V}_{I,j}$ and $V_I$ enter only when converting the best-window risk into an explicit rate relative to the current true CDF. Variation-budget models for nonstationary environments are standard in the nonstationary optimization literature [10]; the variation metric used in this note is specific to lower-orthant conditional CDFs.

9. **Near-oracle dictionary coverage is required only for the simplified rate.** The theorem itself holds for any finite dictionary $\{W_1, \ldots, W_J\}$. The rate

$$O\Big(\sqrt{n_I d V_I} + d V_I\Big)$$

additionally assumes that the dictionary contains a memory length within a constant factor of $W_I^\star$. A geometric dictionary $W_j = 2^{j-1}$ provides this coverage away from the admissible endpoints. This is a design choice, not an assumption on the data-generating process.

10. **Burn-in.** The regret statements are written after the largest required base window is available. This is a bookkeeping condition ensuring that every expert in the dictionary is defined, rather than a stochastic assumption.

**What is not assumed.** In particular, the regret analysis does not require independence, stationarity, mixing, a known drift rate $\rho$, a known variation budget $V_I$, or a confidence level $\delta$. The parameter $\delta$ belongs to the separate high-probability concentration analysis; it does not enter the fixed-share regret or variation-budget oracle inequalities.

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